

## Chapter 5

### Periodic Classification of Elements

1. Who proposed the modern periodic law based on atomic number?

- A) Dobereiner
- B) Mendeleev
- C) Newlands
- D) Moseley

Answer: D) Moseley

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2. In the modern periodic table, elements are arranged according to their:

- A) Atomic mass
- B) Atomic number
- C) Valency
- D) Chemical properties

Answer: B) Atomic number

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3. The vertical columns in the periodic table are called:

- A) Periods
- B) Groups
- C) Series
- D) Rows

Answer: B) Groups

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4. The horizontal rows in the periodic table are called:

- A) Groups
- B) Columns
- C) Periods
- D) Families

Answer: C) Periods

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5. Which of the following elements is an alkali metal?

- A) Calcium
- B) Sodium
- C) Magnesium
- D) Aluminium

Answer: B) Sodium

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6. Which group contains the noble gases?

- A) Group 1
- B) Group 2
- C) Group 17
- D) Group 18

Answer: D) Group 18

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7. Which element has electronic configuration 2, 8, 1?

- A) Neon
- B) Sodium
- C) Magnesium
- D) Aluminium

Answer: B) Sodium

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8. Elements in the same group have:

- A) Same atomic number
- B) Same number of valence electrons
- C) Same atomic mass
- D) Same number of shells

Answer: B) Same number of valence electrons

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9. Which of the following shows a trend of increasing atomic radius?

- A) Left to right in a period
- B) Right to left in a period
- C) Top to bottom in a group
- D) Both B and C

Answer: D) Both B and C

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10. Ionisation energy generally:

- A) Increases down a group and decreases across a period (left to right)
- B) Decreases down a group and increases across a period (left to right)
- C) Remains constant across a period
- D) Is unrelated to position in the table

Answer: B) Decreases down a group and increases across a period (left to right)

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11. Which element is most likely to form a +1 ion?

- A) Oxygen
- B) Chlorine
- C) Potassium

D) Neon

Answer: C) Potassium

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12. Which of these is a halogen?

A) Fluorine

B) Argon

C) Helium

D) Sodium

Answer: A) Fluorine

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13. The periodicity of properties is due to:

A) Repeating atomic masses

B) Repeating electronic configuration after a fixed interval

C) Repeating chemical names

D) Repeating densities

Answer: B) Repeating electronic configuration after a fixed interval

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14. Which of the following is a transition element?

A) Calcium

B) Iron

C) Neon

D) Sodium

Answer: B) Iron

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15. Which of the following has the smallest atomic radius?

A) Li

B) Be

C) B

D) C

Answer: D) C

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16. Which element has the highest electronegativity?

A) Fr (Francium)

B) F (Fluorine)

C) O (Oxygen)

D) Cl (Chlorine)

Answer: B) F (Fluorine)

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17. Which of the following elements is most metallic in character?

- A) Cs (Cesium)
- B) F (Fluorine)
- C) O (Oxygen)
- D) Cl (Chlorine)

Answer: A) Cs (Cesium)

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18. Electronic configuration of an element is 2, 8, 8, 1. The element belongs to:

- A) Group 1, Period 3
- B) Group 1, Period 4
- C) Group 2, Period 4
- D) Group 17, Period 4

Answer: B) Group 1, Period 4

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19. Which group contains elements that are poor conductors of heat and electricity and are insulators in the molten state?

- A) Group 1
- B) Group 2
- C) Group 17
- D) Group 18

Answer: D) Group 18

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20. Which statement is true about atomic size down a group?

- A) It decreases due to increase in nuclear charge
- B) It increases due to addition of new shells
- C) It remains same
- D) It first increases then decreases

Answer: B) It increases due to addition of new shells

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21. Which of the following pairs of elements are in the same period?

- A) Na and K
- B) O and S
- C) Li and Be
- D) F and Cl

Answer: C) Li and Be

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22. Which of these elements has 3 shells and 7 valence electrons?

- A) Chlorine
- B) Phosphorus

C) Sulphur

D) Silicon

Answer: A) Chlorine

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23. The modern periodic table has how many groups?

A) 8

B) 16

C) 18

D) 20

Answer: C) 18

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24. Which of the following is true about metals and non-metals in the periodic table?

A) Metals are on the right side, non-metals on the left

B) Metals are on the left side, non-metals on the right

C) Both are randomly distributed

D) Metals are only in the center

Answer: B) Metals are on the left side, non-metals on the right

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25. Which element would you expect to have properties most similar to magnesium (Mg)?

A) Sodium (Na)

B) Calcium (Ca)

C) Aluminum (Al)

D) Potassium (K)

Answer: B) Calcium (Ca)

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26. Which of the following elements has the configuration  $[\text{Ne}]3s^23p^5$ ?

A) Sulphur

B) Chlorine

C) Argon

D) Phosphorus

Answer: B) Chlorine

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27. Which property decreases from left to right across a period?

A) Electronegativity

B) Ionisation energy

C) Atomic radius

D) Valency

Answer: C) Atomic radius

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28. Noble gases are called inert because:

- A) They have full valence shells and are stable
- B) They are metals
- C) They form many compounds easily
- D) They have high reactivity

Answer: A) They have full valence shells and are stable

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29. Which group contains elements which readily form +2 ions?

- A) Group 1
- B) Group 2
- C) Group 17
- D) Group 18

Answer: B) Group 2

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30. The element with atomic number 12 belongs to which group?

- A) Group 1 (alkali metals)
- B) Group 2 (alkaline earth metals)
- C) Group 13
- D) Group 14

Answer: B) Group 2 (alkaline earth metals)

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